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EATA: Distribution-aware symbolic discovery for explainable trading

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Abstract: Profit-seeking yet explainable trading remains elusive: deep neural predictors excel at accuracy but lack regulatory transparency, whereas symbolic regression yields readable formulas that optimize proxy objectives misaligned with economic utility. We present EATA, a distribution-aware, neural-guided symbolic regression framework that aligns the objective, hypothesis space, and training signal with trading profitability. EATA couples a three-headed policy–value–profit network (PVPNet) with Monte Carlo Tree Search, where accuracy rewards stabilize search while a Wasserstein-derived profit reward captures asymmetric tail risk. A finance-grounded, decoupled grammar with bounded module libraries encodes moving-average, momentum, RSI, volatility, and conditional operators to curb hypothesis-space inflation, while sliding-window inheritance and quantile-consensus signals enable temporal adaptation. Across representative S&P 500 constituent stocks over 2013–2018, EATA delivers an annualized return of 21.2%, a Sharpe ratio of 0.76, and a maximum drawdown of 17%, outperforming strong baselines while maintaining full interpretability; Pareto-frontier analyses confirm a favorable complexity–performance trade-off. Ablation and objective studies show that removing the Wasserstein rewards, neural guidance, or grammar augmentation severely degrades performance, demonstrating that profit-aware symbolic regression can produce explainable trading strategies competitive with black-box counterparts.

Key words: Algorithmic trading, distribution-aware learning, symbolic regression, Wasserstein distance, explainable artificial intelligence, optimal transport

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Pre-print server/online repository

This manuscript has not been previously posted to any preprint server or online repository.

Conflict of interest

The authors declare that they have no conflict of interest.

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